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Frequency estimator based on interpolation of DTFT spectral lines by using maximum sidelobe decay windows

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ABSTRACT

An algorithm for sinusoidal frequency estimation is proposed, which relies on interpolating discrete time Fourier transform (DTFT) spectral lines and utilizing maximum sidelobe decay (MSD) windows. Distinct from the existing approaches, the proposed approach makes use of the DFT peak as well as the DTFT spectral lines at any positions inside the main lobe of the weighted signal spectrum. The mean squared error (MSE) formula is derived. As shown by the simulation results, the presented approach is more approximate to the Cramer-Rao lower bound (CRLB) than the competing approaches with no increase in computational load.

1. Introduction

A vital issue in digital signal processing is sinusoidal frequency estimation, which is involved in communication systems, electronic measurement, biomedicine, radar and sonar [1–5].

Approaches for sinusoidal frequency estimation can be primarily classified as two types: time-domain estimation algorithms [6–12] and frequency-domain estimation algorithms [13–37]. Time-domain methods include maximum likelihood methods [6–8], autocorrelation methods [9–11] and linear prediction methods [12]. Time-domain methods are normally not suitable when high real-time performance is required, as they are computationally intensive. DFT-based approaches stand as the most extensive ones among the frequency-domain approaches. As evaluating a DTFT over a continuous frequency range is expensive, the DFT-based approaches are generally split into two stages. In the first stage, a coarse FFT-based search provides an initial estimate. In the second fine estimation stage, the DFT peak and its adjacent DTFT spectral lines are leveraged to refine the initial estimate. The only difference among existing DFT-based estimators lies in the fine estimation stage. In [13], Yang and Wei harness the DFT peak and its two neighboring DFT bins within the $2N$ -point DFT spectrum following zero-padding for the frequency estimation. Aboutanios and Mulgrew adopt the two DTFT spectral lines sited exactly at the midpoints between the DFT peak and its two adjacent DFT bins [14]. Fang proposes to leverage the DFT bins adjacent to the DFT peak within the $2N$ -point DFT spectrum following zero-padding for fine estimation [19]. Djukanović derives an approach harnessing a curve-fitting method and uses three DTFT

spectral lines to estimate the signal frequency via interpolation [20–21].

The algorithms in [13–27] rely on rectangular window and estimate the frequency in white noise. When there are interference signals besides white noise, non-rectangular windows may be utilized to suppress the influence of interferers on the signal frequency estimation [28–37]. For example, due to nonlinear loads, harmonic interfering signals commonly exist in power systems. In this situation, when the spectral leakage effect of interfering signals prevails over the wideband noise, the spectral leakage can bias the frequency estimates based on rectangular window and non-rectangular windows should be preferable. MSD windows which belong to the cosine class windows and suppress sidelobes at the cost of a wider main lobe and reduced coherent gain, can enhance robustness and suppress spectral leakage at the cost of reduced estimation accuracy. Based on [14,15], Belega and Petri propose the MV-IPDTFT (2) and MV-IPDTFT (3) approach which make use of MSD windows [28]. Candan proposes W-Candan method in [29], which can operate with an arbitrary window function and uses the maximum three DFT bins in the weighted signal spectrum. Presented in [30], the IPDTFT-IR method is based on the MSD windows, which iteratively suppresses the estimation bias caused by the mirror component. In [31], based on the IPDTFT-IR method, the noise-robust algorithm (IPDTFT-NR) is proposed and the frequency estimation performance is improved. In [34], relying on MSD windows the parabolic interpolation based DTFT method is presented to improve the robustness against interference signals. As presented in [32], Fan proposes an interpolated DFT method with MSD windows that harnesses the DFT peak and two symmetrically positioned spectral lines on its both sides with non-fixed

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spacing for frequency estimation. It should be noted that in the absence of interfering signals, nonrectangular windowing deteriorates algorithm accuracy [29].

This paper proposes a sinusoidal frequency estimator based on interpolating DTFT spectral lines and MSD windows. Unlike the competitive algorithms based on MSD windows which harness the spectral lines situated at fixed positions [28,30,31,34,36], the proposed approach possesses greater generality and utilizes the DFT peak and two arbitrarily positioned DTFT spectral lines. Compared with the approach in [32] which utilizes the spectral lines symmetrically distributed on two sides of the DFT peak with any spacing, the proposed approach is also more general because the positions of the used spectral lines are completely arbitrary and do not have to be symmetrically distributed on two sides of the DFT peak. The MSE formula is derived. Simulation experiments are conducted, and the results show that the proposed approach outperforms the competing approaches, and the computational complexity is also scrutinized.

The rest of the paper is constructed as follows. In Section 2, the proposed approach with MSD windows is presented. In Section 3, the MSE formula is derived. In Section 4, the proposed approach is compared with state-of-art approaches and the CRLB. Section 5 summarizes the main conclusions.

2. Proposed estimator

The signal model is.

$$x(n) = s(n) + z(n), n = 0, 1, \dots, N-1 \quad (1)$$

$$s(n) = Ae^{j(2\pi f_0 n / f_s + \theta_0)}, n = 0, 1, \dots, N-1 \quad (2)$$

where $s(n)$ stands for the sinusoid that needs to be estimated. $z(n)$ denotes white noise exhibiting mean of zero and variance of σ^2 . N signifies the sampling numbers. f_0 , A and θ_0 signify the frequency, amplitude and initial phase of the sinusoid, separately. f_s indicates the sampling frequency. The signal-to-noise ratio (SNR) designates the ratio A^2/σ^2 .

We aim to extend the frequency estimator in [24] to the situation when MSD window is used to weight the signal. Classified as cosine class windows, MSD windows are widely utilized in relevant studies [28,30,31,32,33,34,36]. MSD window exhibits a sidelobe decay rate of 6 (2H-1) dB/octave, making it the top performer among all H-term cosine windows [32]. The definition for the MSD window is [32].

$$Q = \frac{|S_w[m+a] \sin\left(\frac{\pi a}{N}\right) \sin(\pi b) - |S_w[m+b] \sin(\pi a) \sin\left(\frac{\pi b}{N}\right)|}{|S_w[m+a] \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) - |S_w[m+b] \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) + |S_w[m] \sin(\pi(a-b))|} \quad (12)$$

$$w(n) = \sum_{h=0}^{H-1} (-1)^h a_h \cos\left(2\pi \frac{h}{N} n\right), n = 0, 1, \dots, N-1 \quad (3)$$

where $H \geq 1$, a_h is the window coefficient and represented as.

$$a_0 = \frac{C^{H-1}}{2^{2H-2}}, a_h = \frac{C^{H-h-1}}{2^{2H-3}}, h = 1, 2, \dots, H-1 \quad (4)$$

The weighted sinusoid using MSD window is.

$$s_w(n) = s(n) \cdot w(n), n = 0, 1, \dots, N-1 \quad (5)$$

For $s_w(n)$, its DTFT is.

$$S_w(\lambda) = AW(\lambda - \nu)e^{j\theta} \quad (6)$$

where $W(\cdot)$ denotes the DTFT of $w(n)$. ν denotes the number of observed signal cycles, and.

$$\nu = m + \delta \quad (7)$$

where m denotes the discrete frequency indexing value corresponding to the DFT peak, while $\delta \in [-0.5, 0.5]$ denotes the discrete frequency deviation from the DFT peak to f_0 .

When $N \gg 1$, the DTFT for $w[n]$ is [28].

$$W(\lambda) = \frac{N \sin(\pi \lambda)}{\pi} \sum_{h=0}^{H-1} (-1)^h a_h \frac{\lambda}{\lambda^2 - h^2} e^{-j\pi \lambda} = \tilde{W}(\lambda) e^{-j\pi \lambda}, n = 0, 1, \dots, N-1 \quad (8)$$

where

$$\tilde{W}(\lambda) = \frac{N \sin(\pi \lambda)}{\pi} \sum_{h=1}^n (-1)^h a_h \frac{\lambda}{\lambda^2 - h^2} \quad (9)$$

After algebraic deductions, the derivative corresponding to $\tilde{W}(\lambda)$ takes the following form [28].

$$\tilde{W}'(\lambda) = \frac{N}{\pi} [\sin(\pi \lambda) + \pi \lambda \cos(\pi \lambda)] \sum_{h=0}^{H-1} \frac{(-1)^h a_h}{\lambda^2 - h^2} - \frac{2N}{\pi} \lambda^2 \sin(\pi \lambda) \sum_{h=0}^{H-1} \frac{(-1)^h a_h}{(\lambda^2 - h^2)^2} \quad (10)$$

The estimation formula (15) in [24] for the discrete frequency deviation δ is.

$$\hat{\delta} = \frac{N}{\pi} \tan^{-1} \left[\frac{|S_a| \sin\left(\frac{\pi a}{N}\right) \sin(\pi b) - |S_b| \sin(\pi a) \sin\left(\frac{\pi b}{N}\right)}{|S_a| \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) - |S_b| \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) + |S_0| \sin(\pi(a-b))} \right] \quad (11)$$

where S_a and S_b are two DTFT samples which lie at $f = (m+a)\Delta f$ and $f = (m+b)\Delta f$ (Δf is the FFT resolution), S_0 is the DFT peak of $s(n)$. (When $\delta \geq 0$, $a, b \in (-0.5, 1)$. When $\delta < 0$, $a, b \in (-1, 0.5)$.)

With the spectral lines in (11) substituted by those in the DTFT spectrum of the weighted signal, the term in the square brackets of (11) is.

where $S_w[m]$ is the maximum DFT sample of $s_w(n)$, $S_w[m+a]$ and $S_w[m+b]$ are two DTFT samples lie at $f = (m+a)\Delta f$ and $f = (m+b)\Delta f$. According to (6), (8) and (9), let $\lambda = m$, $\lambda = m+a$ and $\lambda = m+b$, respectively, we have.

$$|S_w[m]| = |A| \tilde{W}(-\delta) \quad (13)$$

$$|S_w[m+a]| = |A| \tilde{W}(a-\delta) \quad (14)$$

$$|S_w[m+b]| = |A| \tilde{W}(b-\delta) \quad (15)$$

Substituting (13)–(15) into (12) yields.

$$Q = \frac{\tilde{W}(a - \delta) \sin\left(\frac{\pi a}{N}\right) \sin(\pi b) - \tilde{W}(b - \delta) \sin(\pi a) \sin\left(\frac{\pi b}{N}\right)}{\tilde{W}(a - \delta) \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) - \tilde{W}(b - \delta) \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) + \tilde{W}(-\delta) \sin(\pi(a - b))} \quad (16)$$

Conducting first-order Taylor series expansions on $\tilde{W}(-\delta)$, $\tilde{W}(a - \delta)$ and $\tilde{W}(b - \delta)$ around $\lambda = 0$, a and b , respectively, and omitting higher-order terms, we get.

$$\tilde{W}(-\delta) \approx \tilde{W}(0) - \tilde{W}'(0)\delta \quad (17)$$

$$\tilde{W}(a - \delta) \approx \tilde{W}(a) - \tilde{W}'(a)\delta \quad (18)$$

$$\tilde{W}(b - \delta) \approx \tilde{W}(b) - \tilde{W}'(b)\delta \quad (19)$$

By substituting (17)–(19) into (16), we obtain.

$$Q = \frac{\left[\tilde{W}(a) \sin\left(\frac{\pi a}{N}\right) \sin(\pi b) - \tilde{W}(b) \sin(\pi a) \sin\left(\frac{\pi b}{N}\right) \right] - \delta \left[\tilde{W}'(a) \sin\left(\frac{\pi a}{N}\right) \sin(\pi b) - \tilde{W}'(b) \sin(\pi a) \sin\left(\frac{\pi b}{N}\right) \right]}{\left[\tilde{W}(a) \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) - \tilde{W}(b) \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) + \tilde{W}(0) \sin(\pi(a - b)) \right] - \delta \left[\tilde{W}'(a) \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) - \tilde{W}'(b) \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) \right]} \quad (20)$$

Then we define L_1 , L_2 , D_1 and D_2 as follows.

$$L_1 = \tilde{W}(a) \sin\left(\frac{\pi a}{N}\right) \sin(\pi b) - \tilde{W}(b) \sin(\pi a) \sin\left(\frac{\pi b}{N}\right) \quad (21)$$

$$L_2 = \tilde{W}'(a) \sin\left(\frac{\pi a}{N}\right) \sin(\pi b) - \tilde{W}'(b) \sin(\pi a) \sin\left(\frac{\pi b}{N}\right) \quad (22)$$

$$D_1 = \tilde{W}(a) \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) - \tilde{W}(b) \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) + \tilde{W}(0) \sin(\pi(a - b)) \quad (23)$$

$$D_2 = \tilde{W}'(a) \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) - \tilde{W}'(b) \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) \quad (24)$$

From (20)–(24), we get.

$$Q = \frac{L_1 - \delta L_2}{D_1 - \delta D_2} \quad (25)$$

The estimation expression for δ is.

$$\delta = \frac{QD_1 - L_1}{QD_2 - L_2} \quad (26)$$

3. MSE of the proposed method

In the current section, the MSE formula for the proposed algorithm are derived in additive white noise. The observed signal sequence of $S_w(n)$ in (5) is.

$$x_w[n] = s_w[n] + z_w[n], n = 0, 1, \dots, N - 1 \quad (27)$$

where $z_w[n] = z[n] \cdot w[n]$.

Performing N -point DFT on $x_w[n]$, we have.

$$X_w[k] = S_w[k] + Z_w[k], k = 0, 1, \dots, N - 1 \quad (28)$$

DTFT sample positioned at $f = (m + i)\Delta f$ is $X_w[m + i]$, and its amplitude is.

$$|X_w[m + i]| \approx A_{(m+i)\Delta f} + \text{Re}(Z_{(m+i)\Delta f} e^{-j\varphi_{(m+i)\Delta f}}) = A_{(m+i)\Delta f} + U_{(m+i)\Delta f} \quad (29)$$

where $\varphi_{(m+i)\Delta f}$ and $A_{(m+i)\Delta f}$ denote the phase and amplitude of the DTFT of $s_w[n]$ respectively. $U_{(m+i)\Delta f}$ stands as the real part of $Z_{(m+i)\Delta f} e^{-j\varphi_{(m+i)\Delta f}}$.

When $f = m\Delta f$, $(m + a)\Delta f$ and $(m + b)\Delta f$, we have.

$$|X_w[m]| \approx A_{m\Delta f} + U_{m\Delta f} = A_0 + U_0 \quad (30)$$

$$|X_w[m + a]| \approx A_{(m+a)\Delta f} + U_{(m+a)\Delta f} = A_a + U_a \quad (31)$$

$$|X_w[m + b]| \approx A_{(m+b)\Delta f} + U_{(m+b)\Delta f} = A_b + U_b \quad (32)$$

where for the sake of simplicity $A_{m\Delta f}$, $A_{(m+a)\Delta f}$ and $A_{(m+b)\Delta f}$ are represented as A_0 , A_a and A_b respectively. $U_{m\Delta f}$, $U_{(m+a)\Delta f}$ and $U_{(m+b)\Delta f}$ are represented as U_0 , U_a and U_b , respectively.

In the white noise, $|S_w[m]|$, $|S_w[m + a]|$ and $|S_w[m + b]|$ are replaced sequentially with $|X_w[m]|$, $|X_w[m + a]|$ and $|X_w[m + b]|$, yielding the following expression.

$$Q = \frac{|X_w[m + a]| \sin\left(\frac{\pi a}{N}\right) \sin(\pi b) - |X_w[m + b]| \sin(\pi a) \sin\left(\frac{\pi b}{N}\right)}{|X_w[m + a]| \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) - |X_w[m + b]| \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) + |X_w[m]| \sin(\pi(a - b))} \quad (33)$$

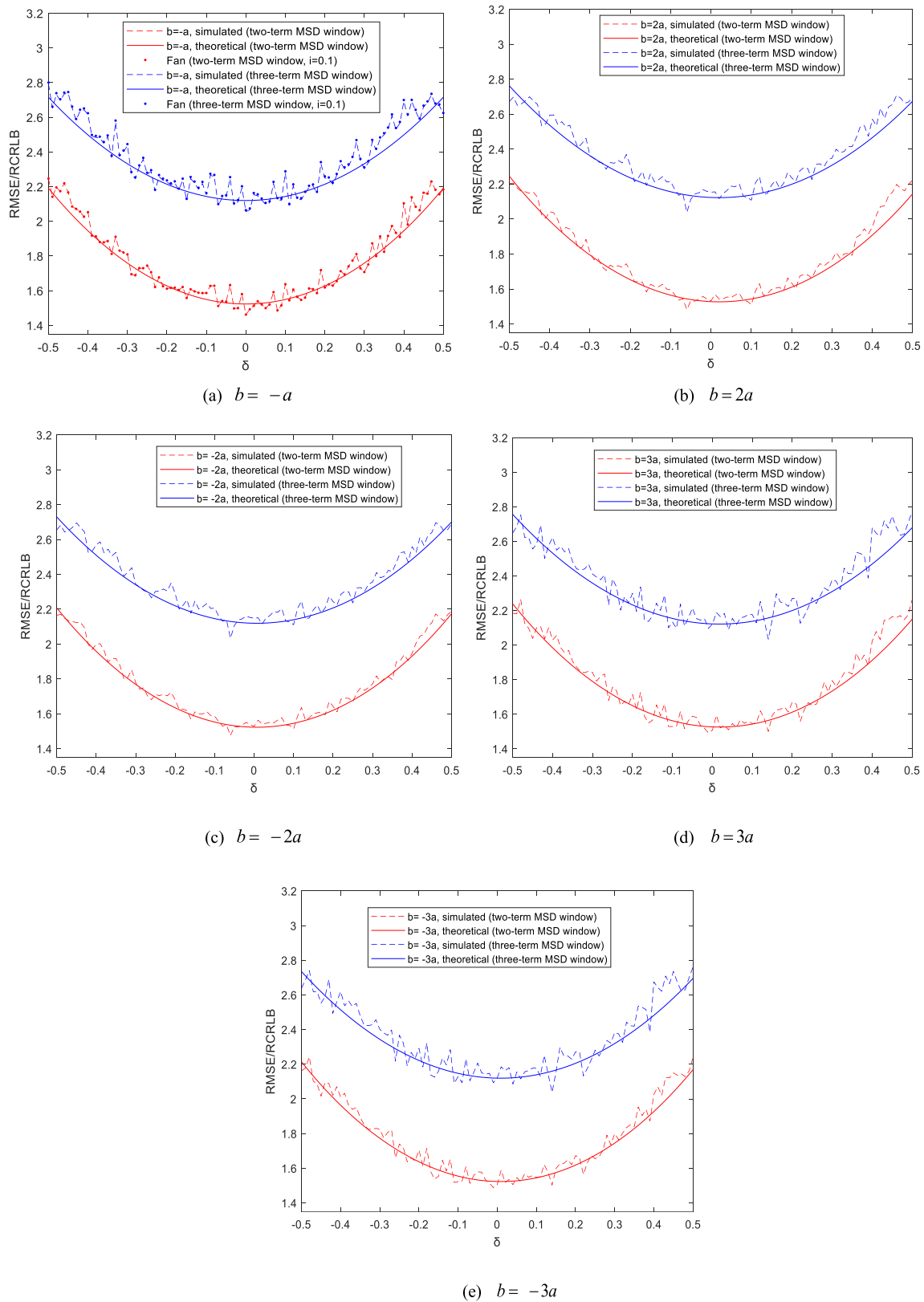


Fig. 1. Theoretical and simulated RMSEs versus δ for the proposed approach.

Then (26) can be written as.

$$\hat{\delta} = \frac{QD_1 - L_1}{QD_2 - L_2} = \frac{Y_1D_1 - Y_2L_1}{Y_1D_2 - Y_2L_2} \quad (34)$$

where the numerator and denominator of Q are denoted by Y_1 and Y_2 sequentially. Substituting (30)–(32) into the numerator of (34), after some algebra we have.

$$\begin{aligned}
 & Y_1D_1 - Y_2L_1 \\
 &= \left(|X_w[m+a]| \sin\left(\frac{\pi a}{N}\right) \sin(\pi b) - |X_w[m+b]| \sin(\pi a) \sin\left(\frac{\pi b}{N}\right) \right) D_1 \\
 &\quad - \left(|X_w[m+a]| \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) - |X_w[m+b]| \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) + |X_w[m]| \sin(\pi(a-b)) \right) L_1 \\
 &= \left[A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_1 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_1 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_1 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_1 \right) \right. \\
 &\quad \left. - A_0 \sin(\pi(a-b)) L_1 \right] + \left[U_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_1 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_1 \right) \right. \\
 &\quad \left. - U_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_1 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_1 \right) - U_0 \sin(\pi(a-b)) L_1 \right]
 \end{aligned} \quad (35)$$

Substituting (30)–(32) into the denominator of (34), after some algebra we have.

$$\begin{aligned}
 & Y_1D_2 - Y_2L_2 \\
 &= \left(|X_w[m+a]| \sin\left(\frac{\pi a}{N}\right) \sin(\pi b) - |X_w[m+b]| \sin(\pi a) \sin\left(\frac{\pi b}{N}\right) \right) D_2 \\
 &\quad - \left(|X_w[m+a]| \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) - |X_w[m+b]| \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) + |X_w[m]| \sin(\pi(a-b)) \right) L_2 \\
 &= \left[A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) \right. \\
 &\quad \left. - A_0 \sin(\pi(a-b)) L_2 \right] + \left[U_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) \right. \\
 &\quad \left. - U_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - U_0 \sin(\pi(a-b)) L_2 \right]
 \end{aligned} \quad (36)$$

Substituting (35) and (36) into (34), after some algebra we can obtain.

$$\hat{\delta} = \frac{A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_1 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_1 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_1 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_1 \right) - A_0 \sin(\pi(a-b)) L_1}{A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - A_0 \sin(\pi(a-b)) L_2} + \frac{U_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_1 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_1 \right) - U_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_1 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_1 \right) - U_0 \sin(\pi(a-b)) L_1}{A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - A_0 \sin(\pi(a-b)) L_2} \times \frac{1}{1 + \frac{U_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - U_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - U_0 \sin(\pi(a-b)) L_2}{A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - A_0 \sin(\pi(a-b)) L_2}} \quad (37)$$

Under high SNR condition, we have.

$$\frac{U_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - U_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_1 \right) - U_0 \sin(\pi(a-b)) L_1}{A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - A_0 \sin(\pi(a-b)) L_2} \ll 1 \quad (38)$$

By performing the first-order Taylor expansion on (37) and ignoring higher-order terms, we obtain.

$$\hat{\delta} \approx \frac{A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_1 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_1 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_1 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_1 \right) - A_0 \sin(\pi(a-b)) L_1}{A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - A_0 \sin(\pi(a-b)) L_2} + \frac{U_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_1 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_1 \right) - U_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_1 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_1 \right) - U_0 \sin(\pi(a-b)) L_1}{A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - A_0 \sin(\pi(a-b)) L_2} + \frac{A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_1 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_1 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_1 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_1 \right) - A_0 \sin(\pi(a-b)) L_1}{A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - A_0 \sin(\pi(a-b)) L_2} + \frac{U_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - U_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - U_0 \sin(\pi(a-b)) L_2}{A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - A_0 \sin(\pi(a-b)) L_2} \times \frac{1}{A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - A_0 \sin(\pi(a-b)) L_2} \quad (39)$$

Substituting $A_0 = S_w[m]$, $A_a = S_w[m + a]$ and $A_b = S_w[m + b]$ into (26), we have.

$$\delta = \frac{A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_1 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_1 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_1 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_1 \right) - A_0 \sin(\pi(a - b)) L_1}{A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - A_0 \sin(\pi(a - b)) L_2} \quad (40)$$

Substituting (40) into (39), we have.

$$\begin{aligned} \hat{\delta} - \delta = & \frac{U_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_1 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_1 - \delta \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) \right)}{A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - A_0 \sin(\pi(a - b)) L_2} \\ & - \frac{U_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_1 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_1 - \delta \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) \right)}{A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - A_0 \sin(\pi(a - b)) L_2} \\ & - \frac{U_0 \sin(\pi(a - b)) (L_1 - \delta L_2)}{A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - A_0 \sin(\pi(a - b)) L_2} \end{aligned} \quad (41)$$

We have $\cos \frac{b\pi}{N} \approx 1$, $\cos \frac{a\pi}{N} \approx 1$, $\sin \frac{b\pi}{N} \approx \frac{b\pi}{N}$ and $\sin \frac{a\pi}{N} \approx \frac{a\pi}{N}$ when $N \gg 1$. We then have.

$$\begin{aligned} & E \left[(\hat{\delta} - \delta)^2 \right] \\ = & \frac{E \left\{ \left[U_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_1 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_1 - \delta \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) \right] \right. \right.}{\left[A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - A_0 \sin(\pi(a - b)) L_2 \right]^2} \\ & \left. \left. - U_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_1 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_1 - \delta \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) \right) \right] \right\}}{\left[A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - A_0 \sin(\pi(a - b)) L_2 \right]^2} \\ & - \frac{U_0 \sin(\pi(a - b)) (L_1 - \delta L_2)^2}{\left[A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - A_0 \sin(\pi(a - b)) L_2 \right]^2} \end{aligned} \quad (42)$$

Then we try to simplify the denominator of (42). As $A_a = A\tilde{W}(a - \delta)$ and $\tilde{W}(\delta)$ has even symmetry, we have.

$$\begin{aligned} & \left[A_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) - A_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) - A_0 \sin(\pi(a - b)) L_2 \right]^2 \\ &= A^2 \left[\tilde{W}(a - \delta) \left(\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \sin(\pi b) L_2 \right) - \tilde{W}(b - \delta) \left(\sin(\pi a) \left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) L_2 \right) - \tilde{W}(\delta) \sin(\pi(a - b)) L_2 \right]^2 \end{aligned} \quad (43)$$

Next, we simplify the numerator of (42) as follows.

$$\begin{aligned} & E \left\{ \left[U_a \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_1 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_1 - \delta \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) \right) \right. \right. \\ & \left. - U_b \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_1 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_1 - \delta \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) \right) - U_0 \sin(\pi(a - b)) (L_1 - \delta L_2) \right]^2 \right\} \\ &= E \left\{ U_a^2 \left[\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_1 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_1 - \delta \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) \right]^2 \right. \\ &+ U_b^2 \left[\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_1 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_1 - \delta \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) \right]^2 + U_0^2 \sin^2(\pi(a - b)) (L_1 - \delta L_2)^2 \\ &- 2U_a U_b \left[\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_1 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_1 - \delta \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) \right] \\ &\quad \times \left[\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_1 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_1 - \delta \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) \right] \\ &- 2U_a U_0 \left[\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_1 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_1 - \delta \left(\sin\left(\frac{\pi a}{N}\right) \sin(\pi b) D_2 - \cos\left(\frac{\pi a}{N}\right) \sin(\pi b) L_2 \right) \right] \sin(\pi(a - b)) (L_1 - \delta L_2) \\ &\left. + 2U_b U_0 \left[\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_1 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_1 - \delta \left(\sin(\pi a) \sin\left(\frac{\pi b}{N}\right) D_2 - \sin(\pi a) \cos\left(\frac{\pi b}{N}\right) L_2 \right) \right] \sin(\pi(a - b)) (L_1 - \delta L_2) \right\} \end{aligned} \quad (44)$$

The autocorrelation function of U_f is [32].

Table 1

The procedures of the presented approach.

Step	Description
1	Weight the sinusoidal signal using MSD window, and acquire $x_w[n] = x[n] \cdot w[n]$, $n = 0, 1, \dots, N - 1$.
2	N -point DFT is executed on $x_w[n]$
3	Compute the difference between $X_{w(1)}$ and $X_{w(-1)}$, and get $q_w = X_{w(1)} - X_{w(-1)} $
4	If $q_w \geq 0$
5	Compute $X_{w(0.25)}$ and $X_{w(0.25)}$ by $X_{w(C_w)} = \sum_{n=0}^{N-1} x_w(n) e^{-j2\pi n \frac{\tilde{m}_w + C_w}{N}}$, $C_w = 0.25, 0.5$
6	Acquire the preliminary estimate $\hat{\delta}_1$ utilizing $X_{w(0)}$, $X_{w(0.25)}$ and $X_{w(0.5)}$ ($a = 0.25, b = 0.5$) by (26) and (12)
7	Find the maximum value among $ X_{w(0)} $, $ X_{w(0.25)} $, $ X_{w(0.5)} $ and get p_1
8	Compute $X_{w(\hat{\delta}_1)}$ and $X_{w(2\hat{\delta}_1 - p_1)}$ by $X_{w(E_w)} = \sum_{n=0}^{N-1} x_w(n) e^{-j2\pi n \frac{\tilde{m}_w + E_w}{N}}$, $E_w = \hat{\delta}_1, 2\hat{\delta}_1 - p_1$
9	Calculate the fine estimate $\hat{\delta}_2$ ($a = p_1 - \hat{\delta}_1, b = \hat{\delta}_1 - p_1$) with $X_{w(\hat{\delta}_1)}$, $X_{w(p_1)}$ and $X_{w(2\hat{\delta}_1 - p_1)}$ according to (26) and (12)
10	The final estimation value $\hat{\delta}_f = \hat{\delta}_1 + \hat{\delta}_2$
11	Else
12	Compute $X_{w(-0.25)}$ and $X_{w(-0.5)}$ by $X_{w(D_w)} = \sum_{n=0}^{N-1} x_w(n) e^{-j2\pi n \frac{\tilde{m}_w + D_w}{N}}$, $D_w = -0.25, -0.5$
13	Acquire the preliminary estimate $\hat{\delta}_1$ utilizing $X_{w(0)}$, $X_{w(-0.25)}$ and $X_{w(-0.5)}$ ($a = -0.25, b = -0.5$) by (26) and (12)
14	Find the maximum value among $ X_{w(0)} $, $ X_{w(-0.25)} $, $ X_{w(-0.5)} $, and get p_2
15	Compute $X_{w(\hat{\delta}_1)}$ and $X_{w(2\hat{\delta}_1 - p_2)}$ by $X_{w(F_w)} = \sum_{n=0}^{N-1} x_w(n) e^{-j2\pi n \frac{\tilde{m}_w + F_w}{N}}$, $F_w = \hat{\delta}_1, 2\hat{\delta}_1 - p_2$
16	Calculate the fine estimate $\hat{\delta}_2$ ($a = p_2 - \hat{\delta}_1, b = \hat{\delta}_1 - p_2$) with $X_{w(\hat{\delta}_1)}$, $X_{w(p_2)}$ and $X_{w(2\hat{\delta}_1 - p_2)}$ by (26) and (12)
17	The final estimation value $\hat{\delta}_f = \hat{\delta}_1 + \hat{\delta}_2$

$$E(U_{f_1}, U_{f_2}) = \frac{\sigma^2}{2} |W_2(f_1 - f_2)| = \frac{\sigma^2}{2} \sum_{n=0}^{N-1} w^2(n) e^{-j2\pi(f_1 - f_2)Tn/N} \quad (45)$$

where T is the signal observation time length.

We have.

$$E(U_a U_0) = \frac{\sigma^2}{2} |W_2(a\Delta f)| \quad (46)$$

$$E(U_a U_0) = \frac{\sigma^2}{2} |W_2((a - b)\Delta f)| \quad (47)$$

$$E(U_b U_0) = \frac{\sigma^2}{2} |W_2(b\Delta f)| \quad (48)$$

$$E(U_0^2) = E(U_a^2) = E(U_b^2) = \frac{\sigma^2}{2} |W_2(0)| \quad (49)$$

The MSE of $\hat{\delta}$ is.

$$\begin{aligned} & E[(\hat{\delta} - \delta)^2] \\ & \approx \frac{|W_2(0)| \left[\sin^2(\pi b) \left(\sin\left(\frac{\pi a}{N}\right) D_1 - \cos\left(\frac{\pi a}{N}\right) L_1 - \delta \left(\sin\left(\frac{\pi a}{N}\right) D_2 - \cos\left(\frac{\pi a}{N}\right) L_2 \right) \right)^2 \right]}{2SNR \left[\tilde{W}(a - \delta) \left(\left(\frac{\pi a}{N} \right) \sin(\pi b) D_2 - \sin(\pi b) L_2 \right) - \tilde{W}(b - \delta) \left(\sin(\pi a) \left(\frac{\pi b}{N} \right) D_2 - \sin(\pi a) L_2 \right) - \tilde{W}(\delta) \sin(\pi(a - b)) L_2 \right]^2} \\ & + \frac{\sin^2(\pi a) \left(\sin\left(\frac{\pi b}{N}\right) D_1 - \cos\left(\frac{\pi b}{N}\right) L_1 - \delta \left(\sin\left(\frac{\pi b}{N}\right) D_2 - \cos\left(\frac{\pi b}{N}\right) L_2 \right) \right)^2 + \sin^2(\pi(a - b)) (L_1 - \delta L_2)^2}{2SNR \left[\tilde{W}(a - \delta) \left(\left(\frac{\pi a}{N} \right) \sin(\pi b) D_2 - \sin(\pi b) L_2 \right) - \tilde{W}(b - \delta) \left(\sin(\pi a) \left(\frac{\pi b}{N} \right) D_2 - \sin(\pi a) L_2 \right) - \tilde{W}(\delta) \sin(\pi(a - b)) L_2 \right]^2} \\ & + \frac{|W_2((a - b)\Delta f)| \sin(\pi a) \sin(\pi b) \left(\sin\left(\frac{\pi a}{N}\right) D_1 - \cos\left(\frac{\pi a}{N}\right) L_1 - \delta \left(\sin\left(\frac{\pi a}{N}\right) D_2 - \cos\left(\frac{\pi a}{N}\right) L_2 \right) \right)}{SNR \left[\tilde{W}(a - \delta) \left(\left(\frac{\pi a}{N} \right) \sin(\pi b) D_2 - \sin(\pi b) L_2 \right) - \tilde{W}(b - \delta) \left(\sin(\pi a) \left(\frac{\pi b}{N} \right) D_2 - \sin(\pi a) L_2 \right) - \tilde{W}(\delta) \sin(\pi(a - b)) L_2 \right]^2} \\ & \times \frac{\left(\sin\left(\frac{\pi b}{N}\right) D_1 - \cos\left(\frac{\pi b}{N}\right) L_1 - \delta \left(\sin\left(\frac{\pi b}{N}\right) D_2 - \cos\left(\frac{\pi b}{N}\right) L_2 \right) \right)}{SNR \left[\tilde{W}(a - \delta) \left(\left(\frac{\pi a}{N} \right) \sin(\pi b) D_2 - \sin(\pi b) L_2 \right) - \tilde{W}(b - \delta) \left(\sin(\pi a) \left(\frac{\pi b}{N} \right) D_2 - \sin(\pi a) L_2 \right) - \tilde{W}(\delta) \sin(\pi(a - b)) L_2 \right]^2} \\ & + \frac{|W_2(a\Delta f)| \sin(\pi b) \sin(\pi(a - b)) (L_1 - \delta L_2) \left(\sin\left(\frac{\pi a}{N}\right) D_1 - \cos\left(\frac{\pi a}{N}\right) L_1 - \delta \left(\sin\left(\frac{\pi a}{N}\right) D_2 - \cos\left(\frac{\pi a}{N}\right) L_2 \right) \right)}{SNR \left[\tilde{W}(a - \delta) \left(\left(\frac{\pi a}{N} \right) \sin(\pi b) D_2 - \sin(\pi b) L_2 \right) - \tilde{W}(b - \delta) \left(\sin(\pi a) \left(\frac{\pi b}{N} \right) D_2 - \sin(\pi a) L_2 \right) - \tilde{W}(\delta) \sin(\pi(a - b)) L_2 \right]^2} \\ & + \frac{|W_2(b\Delta f)| \sin(\pi a) \sin(\pi(a - b)) (L_1 - \delta L_2) \left(\sin\left(\frac{\pi b}{N}\right) D_1 - \cos\left(\frac{\pi b}{N}\right) L_1 - \delta \left(\sin\left(\frac{\pi b}{N}\right) D_2 - \cos\left(\frac{\pi b}{N}\right) L_2 \right) \right)}{SNR \left[\tilde{W}(a - \delta) \left(\left(\frac{\pi a}{N} \right) \sin(\pi b) D_2 - \sin(\pi b) L_2 \right) - \tilde{W}(b - \delta) \left(\sin(\pi a) \left(\frac{\pi b}{N} \right) D_2 - \sin(\pi a) L_2 \right) - \tilde{W}(\delta) \sin(\pi(a - b)) L_2 \right]^2} \end{aligned} \quad (50)$$

Then we define.

$$C_1 = \sin(\pi a/N) D_1 - \cos(\pi a/N) L_1 \quad (51)$$

$$C_2 = \sin(\pi a/N) D_2 - \cos(\pi a/N) L_2 \quad (52)$$

Table 2

Parameters used for simulation.

Parameters	Definitions
A	Amplitude of the sinusoid
θ_0	Phase of the sinusoid
f_0	Frequency of the sinusoid
δ	Discrete frequency deviation
N	Sampling numbers
v	Number of observed signal cycles
SNR	Signal-to-noise ratio
SIR	Signal-to-interference ratio
THD	Total Harmonic Distortion
Δf	FFT resolution

$$C_3 = (\pi a/N) \sin(\pi b) D_2 - \sin(\pi b) L_2 \quad (53)$$

$$C_4 = (\pi b/N) \sin(\pi a) D_2 - \sin(\pi a) L_2 \quad (54)$$

$$C_5 = \sin(\pi b/N) D_1 - \cos(\pi b/N) L_1 \quad (55)$$

$$C_6 = \sin(\pi b/N) D_2 - \cos(\pi b/N) L_2 \quad (56)$$

$$C_7 = \sin(\pi(a - b)) (L_1 - \delta L_2) \quad (57)$$

$$C_8 = \sin(\pi(a - b)) L_2 \quad (58)$$

Substituting (51)–(58) into (50), we can obtain.

$$\begin{aligned}
 & E[(\hat{\delta} - \delta)^2] \\
 & \approx \frac{|W_2(0)| [\sin^2(\pi b)(C_1 - \delta C_2)^2 + \sin^2(\pi a)(C_5 - \delta C_6)^2 + C_7^2]}{2\text{SNR} [\tilde{W}(a - \delta)C_3 - \tilde{W}(b - \delta)C_4 - \tilde{W}(\delta)C_8]^2} - \frac{|W_2(a - b)\Delta f| \sin(\pi a) \sin(\pi b) (C_1 - \delta C_2)(C_5 - \delta C_6)}{\text{SNR}^2 [\tilde{W}(a - \delta)C_3 - \tilde{W}(b - \delta)C_4 - \tilde{W}(\delta)C_8]^4} \\
 & \quad - \frac{|W_2(a\Delta f)| \sin(\pi b) C_7 (C_1 - \delta C_2) - |W_2(b\Delta f)| \sin(\pi a) C_7 (C_5 - \delta C_6)}{\text{SNR} [\tilde{W}(a - \delta)C_3 - \tilde{W}(b - \delta)C_4 - \tilde{W}(\delta)C_8]^2}
 \end{aligned} \tag{59}$$

Fig. 1 presents the variation of simulated and theoretical RMSEs versus the discrete frequency deviation δ for the proposed approach, with N set to 16 and SNR fixed at 10 dB. The RMSEs are normalized to the square root of CRLB (RCRLB). The relationships between b and a in Fig. 1(a)–(e) are $b = -a$, $b = 2a$, $b = -2a$, $b = 3a$ and $b = -3a$, respectively. According to the proposed approach, when $\delta \geq 0$, $a, b \in (-0.5, 1)$, and when $\delta < 0$, $a, b \in (-1, 0.5)$. a and b can take any values within the above ranges. In the simulation experiment of Fig. 1, we only take $b = -a$, $b = 2a$, $b = -2a$, $b = 3a$ and $b = -3a$ as examples to verify the correctness of the derived MSE formula. When other values or relationships are used, the same conclusion can be drawn. We obtain the theoretical RMSE curves from (59). The simulated RMSE curves are obtained via (26) and (12). δ varies from -0.5 to 0.5 with a step of 0.01 . Fan method is used in the first iteration. We use $b = 0.1$ in these figures. For each value of δ , 1000 independent experimental trials are executed. In cases where the frequency, amplitude and initial phase of the complex sinusoid are unknown, the CRLB is [6].

$$\text{CRLB} = \min_{\hat{\delta}} \frac{3N}{2\pi^2(N^2 - 1)\text{SNR}} \tag{60}$$

Fig. 1(a)–(e) verifies that the curves of the simulation match well with the theoretical curves. It can additionally be inferred from Fig. 1(a) that when $b = -a$, Fan method ($i = 0.1$) achieves very similar performance with the proposed approach. Therefore, Fan approach is a particular instance of the proposed approach when $b = -a$.

Table 1 contains the complete steps of the proposed approach. The

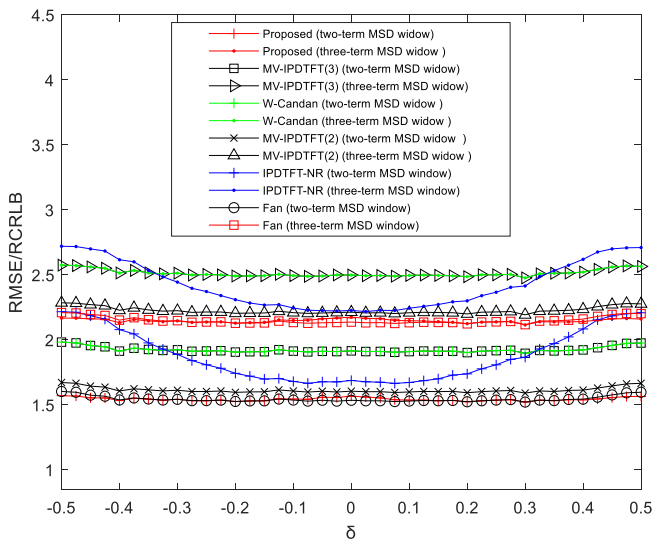


Fig. 2. Simulated RMSEs versus the discrete frequency deviation δ .

sinusoid $x(n)$ is weighted using MSD window, and $x_w(n)$ is obtained. The DFT of $x_w(n)$, represented by $X_w[m + k]$, is denoted as $X_{w(k)}$ for brevity. Then q_w (defined as the difference between $|X_{w(1)}|$ and $|X_{w(-1)}|$) is obtained. Supposing that $q_w \geq 0$ (which means $0 \leq \delta \leq 0.5$), the signal frequency is situated between $X_{w(0)}$ and $X_{w(0.5)}$. Hence, we harness $X_{w(0)}$, $X_{w(0.25)}$ and $X_{w(0.5)}$ to calculate a preliminary estimate $\hat{\delta}_1$ in step 6. p_1 is ascertained by extracting the subscript of the maximum value among $|X_{w(0)}|$, $|X_{w(0.25)}|$ and $|X_{w(0.5)}|$. This is done because the spectral lines with greater amplitudes lie nearer to the signal frequency, and thus are less susceptible to the noise. During step 9, $X_{w(\hat{\delta}_1)}$, $X_{w(p_1)}$ and $X_{w(2\hat{\delta}_1 - p_1)}$ are harnessed to obtain the fine estimation $\hat{\delta}_2$. $X_{w(p_1)}$ and $X_{w(2\hat{\delta}_1 - p_1)}$ lie evenly on the two sides of $X_{w(\hat{\delta}_1)}$, and they are with comparatively greater amplitudes. In step 10, we obtain the final estimation value as $\hat{\delta}_f = \hat{\delta}_1 + \hat{\delta}_2$. Here we use similar iterative ideas as the approaches in [28,29] and [32], and change the operating point of the proposed estimator from large $|\delta|$ to small $|\delta|$ for improving the estimation performance for all the values of δ . For the values of q_w satisfying $q_w < 0$, we utilize similar steps. In step 13, we leverage $X_{w(0)}$, $X_{w(-0.25)}$ and $X_{w(-0.5)}$ to gain the preliminary estimate $\hat{\delta}_1$. p_2 is acquired from the subscript corresponding to the maximum among $|X_{w(0)}|$, $|X_{w(-0.25)}|$ and $|X_{w(-0.5)}|$ in step 14. During step 16, we acquire the fine estimate $\hat{\delta}_2$ by employing $X_{w(\hat{\delta}_1)}$, $X_{w(p_2)}$ and $X_{w(2\hat{\delta}_1 - p_2)}$ via (26) and (12). In step 17, we use the same idea as step 10 to obtain the final estimate as $\hat{\delta}_f = \hat{\delta}_1 + \hat{\delta}_2$.

4. Simulation experiments

In this section, we carry out performance comparison between the proposed approach and the comparative approaches and undertake the analysis of the simulation results. The approaches used for comparison include MV-IPDTFT (2) [28], MV-IPDTFT (3) [28], W-Candan [29], IPDTFT-NR [31] and Fan [32] approach. For the Fan method, the interval between the spectral lines is set as 0.1 . The experiments are delineated over three distinct cases: noisy sinusoid, noisy sinusoid affected by single-tone interference signal and noisy and harmonically distorted sinusoid. For the sinusoidal signal, the amplitude is set to $A = 1$, the frequency is $f_0 = \left(\frac{N}{4} + \delta\right) \frac{f_s}{N}$, its initial phase θ_0 is evenly distributed in the range $[0, 2\pi]$. The parameters used for simulation are listed in Table 2.

4.1. Noisy sinusoid

This part we analyze the performance of distinct approaches in additive white noise. Fig. 2 demonstrates the simulated RMSEs (which are normalized to the RCRLB) versus δ for the proposed, IPDTFT-NR, W-Candan, Fan, MV-IPDTFT (2) and MV-IPDTFT (3) approach, with N is set to 16 and SNR is equal to 20 dB. $\delta \in [-0.5, 0.5]$ and its step is 0.025 .

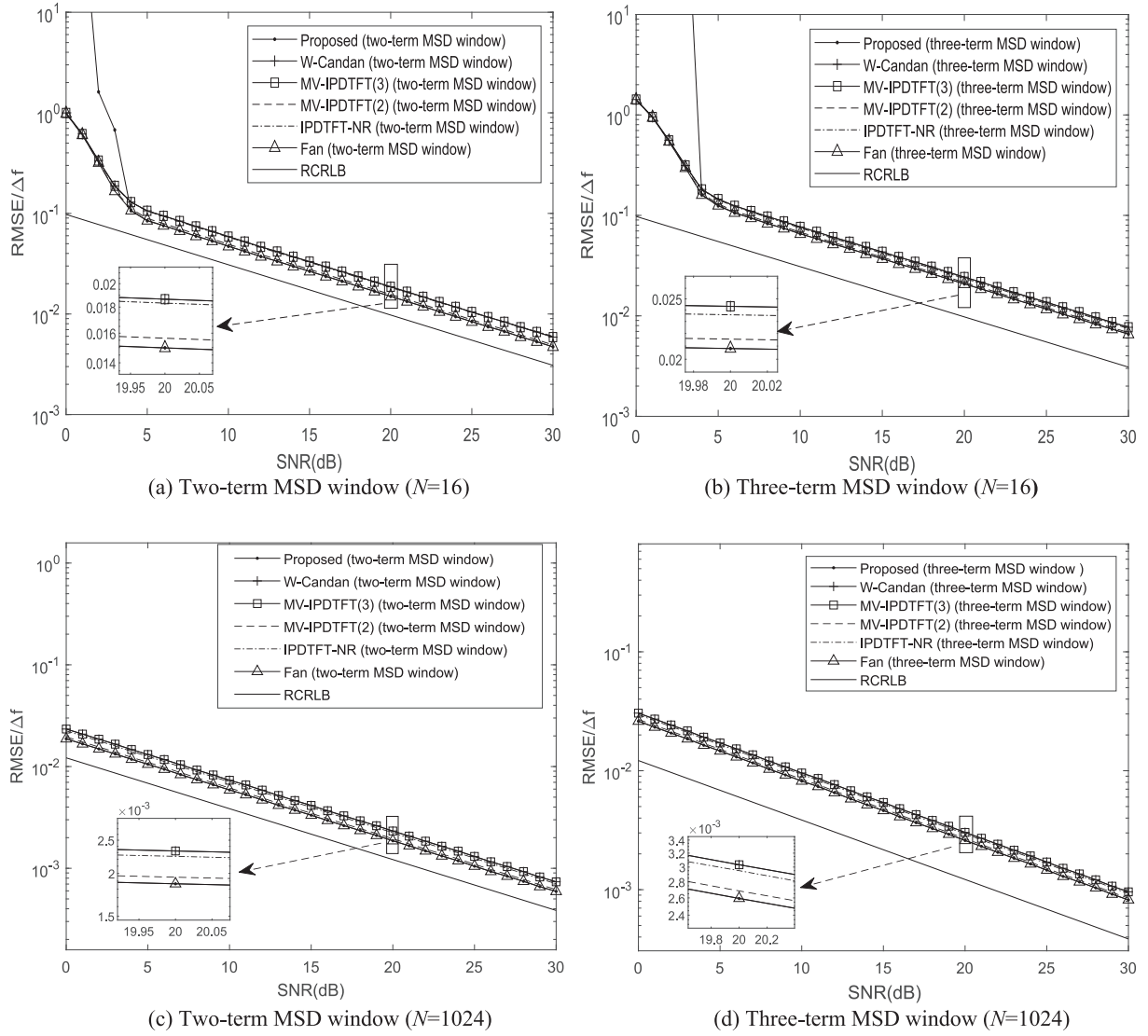


Fig. 3. Simulated RMSEs for distinct approaches versus SNR.

Implementation of 20,000 runs is carried out for every individual value of δ . As depicted in this figure, the proposed and the Fan approach have very similar performance when the same MSD window is used. When two-term MSD window is adopted, these two methods exhibit superior performance to the other approaches. With respect to the identical approach, the estimation accuracy using the two-term MSD window outperforms that using the three-term MSD window. The RMSE curves corresponding to the MV-IPDTFT (3) and the W-Candan approach display very similar performance.

For $N = 16$ and $\delta = 0.3$, Fig. 3(a) and (b) depict the simulated RMSEs (normalized to Δf) versus SNR for distinct approaches. We perform a total of 50,000 runs for each distinct SNR value. The performance of the Fan method is very close to that of the proposed method, and these two methods align more closely with the RCRLB and outperform the other approaches in the majority of cases. When $N = 1024$ and other parameters remain the same, Fig. 3(c) and (d) show the simulated RMSEs versus SNR for different methods. The performance of the Fan method is very close to that of the proposed method when the same MSD window is used, and they are closer to the RCRLB than the other methods.

4.2. Noisy sinusoid affected by single-tone interference signal

Relative to the sinusoidal frequency that is to be estimated, the fre-

quency of the single-tone interference sinusoid is supposed to be $6\Delta f$ higher, with the SIR (signal-to-interference ratio) valued at 3 dB. Both the initial phase of the sinusoid and the initial phase of interference signal exhibit a uniform distribution within the range $[0, 2\pi]$.

Fig. 4(a) and (b) illustrate the RMSE (normalized to Δf) of the proposed approach and the competing approaches versus SNR when $N = 16$ and $\delta = 0.3$. The proposed and the competing methods utilizing two-term and three-term MSD window. 50,000 experimental runs are conducted for each SNR value. These figures show that the Fan method has similar performance to the proposed method. When narrowband interference signal exists, the proposed and Fan method achieve superior accuracy compared with the other methods. When $N = 1024$ and other parameters remain the same, Fig. 4(c) and (d) show the RMSE of different approaches versus SNR when $N = 1024$ and $\delta = 0.3$. It can be seen from Fig. 4(c) that when two-term MSD window is used, the proposed and Fan method have similar performance, and they perform better than the other methods when the SNR is lower than 15 dB. When the SNR is higher than 15 dB, the MV-IPDTFT (3) and W-Candan method with two-term MSD window perform better than the other methods. From Fig. 4(d), we can see that when three-term MSD window is used, the proposed and Fan method have similar performance, and they are closer to the RCRLB than the other methods.

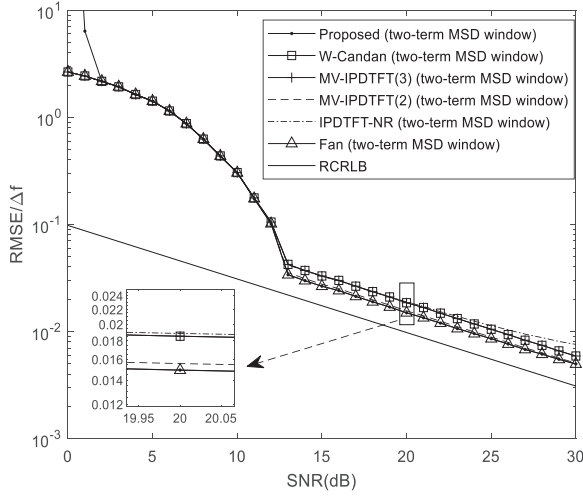
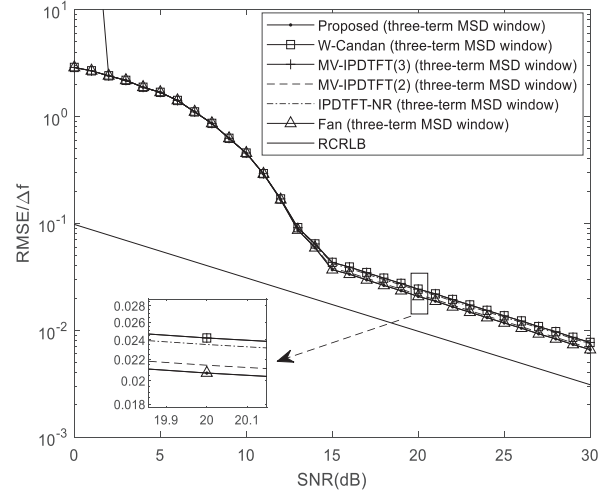
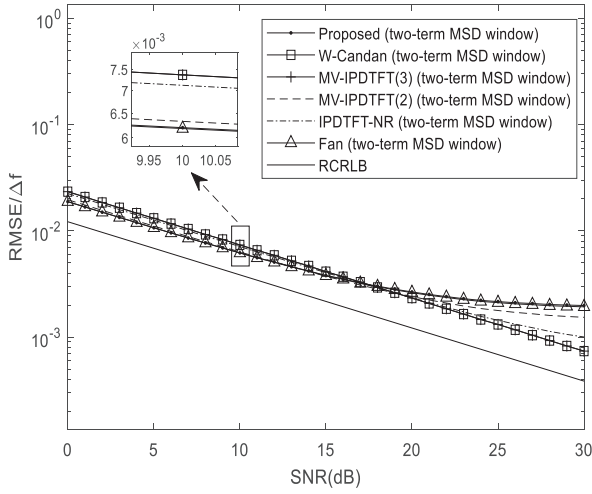
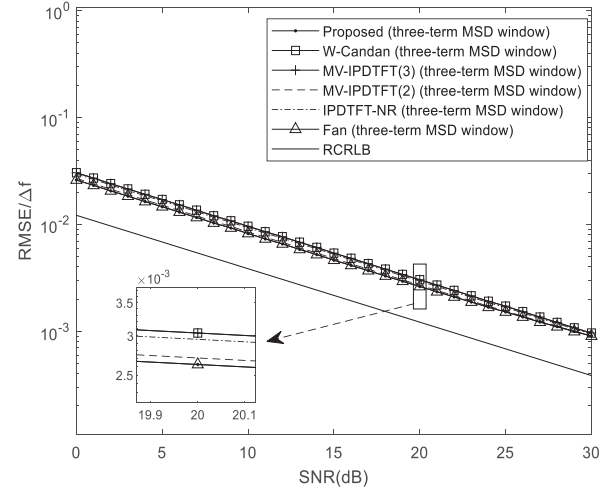
(a) Two-term MSD window ($N=16$)(b) Three-term MSD window ($N=16$)(c) Two-term MSD window ($N=1024$)(d) Three-term MSD window ($N=1024$)

Fig. 4. Simulated RMSEs for distinct approaches versus SNR.

4.3. Noisy and harmonically distorted sinusoid

The 2nd, 3rd, and 4th harmonics with an amplitude ratio of 8:4:1 are hypothesized to exist for the subsequent experimental analysis. The THD (Total Harmonic Distortion) is specified as 60%. Both the initial phase of the sinusoid and the initial phases of the harmonics exhibit a uniform distribution within the limits of $[0, 2\pi]$. We adopt a step size of $1/12$ for the number of observed signal cycles ν which ranges from 2.54 to 12.04. 20,000 experimental trials are implemented for each ν value. We set N as 512 and SNR as 20 dB.

The simulated MSEs with respect to ν corresponding to the proposed and MV-IPDTFT (3) approach are presented in Fig. 5(a). It is demonstrated that the proposed approach relies on two-term MSD window has superior performance to competing approaches in two conditions: when $\nu > 5.19$ or $\nu < 5.19$ and $|\delta|$ tends to 0.5. The MV-IPDTFT (3) approach employing the two-term MSD window outperforms the rival approaches when $\nu < 5.19$ and $|\delta|$ tends to 0.

The simulated MSEs of the proposed approach as well as the IPDTFT-NR approach, with respect to ν are illustrated in Fig. 5(b). In the case of $\nu > 6.12$ or $\nu < 6.12$ and $|\delta|$ tends to 0.5, the proposed approach relying on the two-term MSD window has superior performance compared with the competing approaches in the majority of instances. The IPDTFT-NR

approach that relies on the two-term MSD window outperforms other approaches in instances where $\nu < 6.12$ and $|\delta|$ tends to 0.

In Fig. 5(c), the MSE values for the proposed and W-Candan approach with respect to ν are illustrated. It is concluded that the proposed approach which employs the two-term MSD window achieves the minimum MSE among all approaches in the cases of $\nu > 5.19$ or $\nu < 5.19$ and $|\delta|$ tends to 0.5. For cases where $\nu < 5.19$ and $|\delta|$ tends to 0, the W-Candan approach relies on the two-term MSD window exhibits superior performance compared with other approaches.

The simulated MSEs with respect to ν of the proposed approach as well as the MV-IPDTFT (2) approach are presented in Fig. 5(d). In instances where $\nu > 6.12$, the proposed approach relies on the two-term MSD window outperforms all competitors. When $4.28 < \nu < 6.12$, both the proposed and the MV-IPDTFT (2) approach which employ the two-term MSD window exhibit superior performance relative to competing approaches in most circumstances. The MV-IPDTFT (2) approach that employs the two-term MSD window delivers performance superiority over the other approaches in the range of $\nu < 4.28$.

The simulated MSEs with respect to ν of the proposed approach and the Fan approach are presented in Fig. 5(e). In the whole value range of ν , the performance of the Fan and the presented method are very close to each other when the same window function is used.

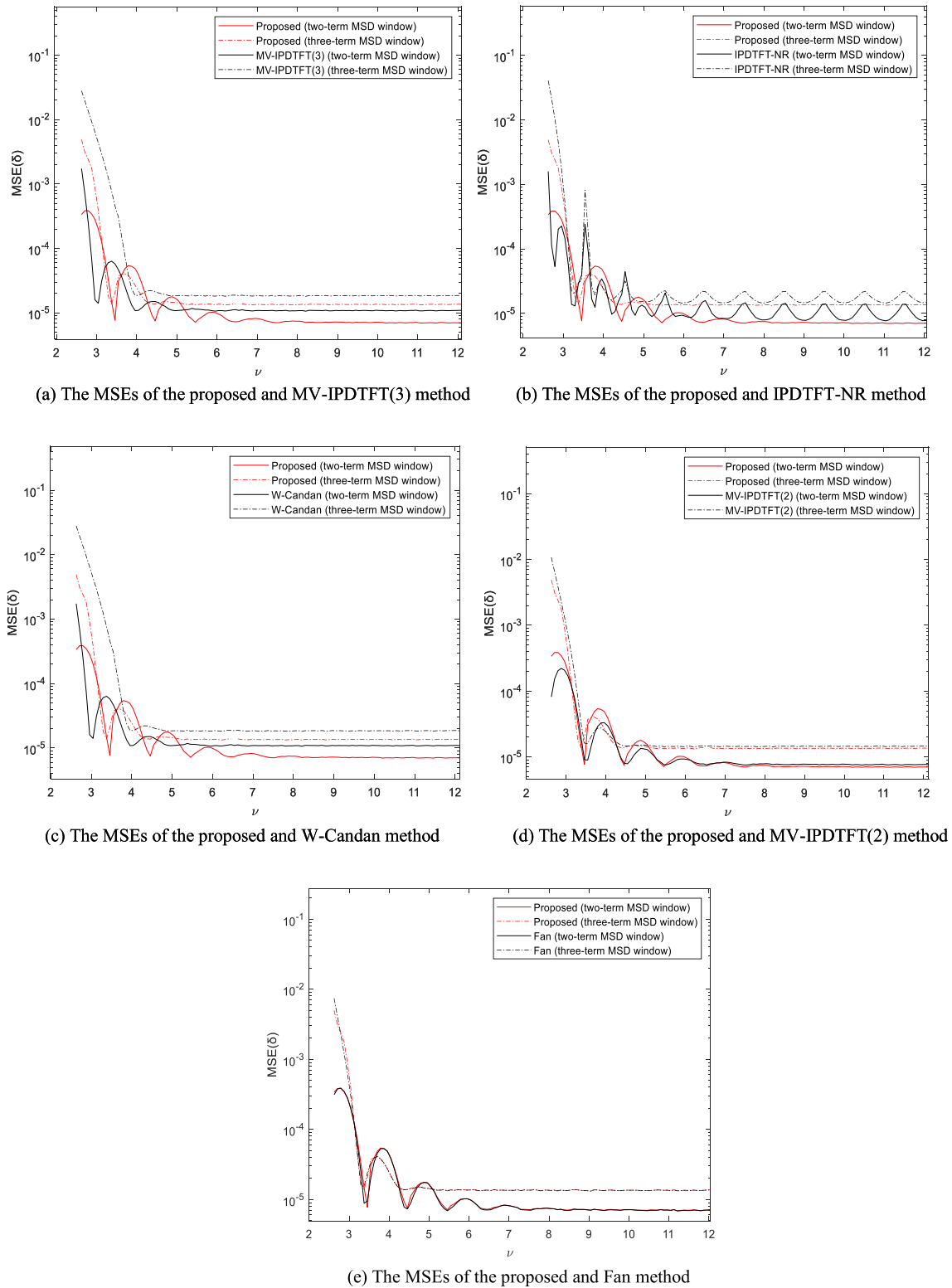


Fig. 5. Simulated MSEs versus ν for distinct approaches.

The computational requirements corresponding to distinct algorithms are listed in Table 3, with a single N -point FFT demanding $(N/2)\log_2 N$ complex multiplications and $N\log_2 N$ complex additions. In step 5 or 12 of Table 1, the proposed approach attains two DTFT samples and alongside two DTFT samples in step 8 or 15. Hence, the proposed approach entails four DTFT samples needing $4N$ complex multiplications and $(4N-4)$ complex additions. The overall computational cost of

the proposed approach comprises $[(N/2)\log_2 N + 4N + 2]$ complex multiplications and $(N\log_2 N + 4N - 4)$ complex additions. When $N = 1024$, the complex multiplications executed by the proposed, MV-IPDTFT (3), IPDTFT-NR, W-Candan, MV-IPDTFT (2) and Fan algorithm are quantified as 9216, 8194, 13313, 8194, 9216 and 10,240 respectively. Thus, in terms of computational complexity, the proposed approach is equivalent to MV-IPDTFT (2) approach and lower than

Table 3
Computational requirements of distinct approaches.

Approach	Complex multiplications	Complex additions
Presented	$\frac{N}{2}\log_2 N + 4N$	$N\log_2 N + 4N - 4$
W-Candan	$\frac{N}{2}\log_2 N + 3N + 2$	$N\log_2 N + 3N + 3$
MV-IPDTFT (3)	$\frac{N}{2}\log_2 N + 3N + 2$	$N\log_2 N + 3N + 3$
MV-IPDTFT (2)	$\frac{N}{2}\log_2 N + 4N$	$N\log_2 N + 4N - 4$
IPDTFT-NR	$\frac{N}{2}\log_2 N + 8N + 1$	$N\log_2 N + 8N - 6$
Fan	$\frac{N}{2}\log_2 N + 5N$	$N\log_2 N + 5N - 5$

IPDTFT-NR and Fan approach. The computational loads of MV-IPDTFT (3) and W-Candan approach are lower than those exhibited by the proposed approach. Nevertheless, MV-IPDTFT (3) and W-Candan algorithm have lower accuracy than the proposed approach which is shown by the results of Figs. 2–4. From the simulation results of Figs. 4 and 5 and the analysis of Table 3, we can see that in the case of noisy sinusoid affected by single-tone interference signal or noisy and harmonically distorted sinusoid, the presented approach achieves lower frequency estimation error without increasing the computational complexity compared with the existing DFT-based windowing approaches. Therefore, the presented approach can effectively suppress the detrimental influence of interference signals and is suitable to be used in harmonic analysis of power systems or liquid level measurement with linear frequency modulated continuous wave radar.

5. Conclusion

A sinusoidal frequency estimation approach is presented in this article which harnesses MSD windows and is based on interpolation of DTFT spectral lines. To suppress the detrimental effect of interference signals on frequency estimation, the sinusoidal signal is weighted using MSD windows. The DFT peak and two arbitrarily positioned DTFT spectral lines inside the main lobe of the weighted signal spectrum are employed in the fine estimation. The MSE formula is deduced in additive white noise. As demonstrated by the simulation results, the Fan approach can be considered as a specific case of the presented approach ($b = -a$). In the cases of noisy sinusoid and noisy sinusoid affected by single-tone interference signal, when the SNR varies from 0 to 30 dB and N equals 16 or 1024, the proposed and the Fan approach have very similar performance when the same MSD window is used, and they perform better than the other MSD windowing approaches in most situations. In the case of noisy and harmonically distorted sinusoid, the proposed approach also outperforms MV-IPDTFT (3), IPDTFT-NR, W-Candan and MV-IPDTFT (2) approach. Its computational complexity is almost identical to that of MV-IPDTFT (2) approach and is lower than that of Fan and IPDTFT-NR approach. The proposed approach exhibits good generality and can be advantageously used in practice when the spectral leakage effect of interfering signals prevails over the wideband noise. For future work, we aim to extend the proposed method by incorporating deep learning techniques to deliver superior frequency estimation performance.

CRedit authorship contribution statement

Lanxin Lu: Writing – original draft, Software, Conceptualization. **Mingjie Li:** Formal analysis, Data curation. **Yanxu Hou:** Validation, Project administration. **XinRui Zhao:** Validation, Resources. **Jiyu Jin:** Methodology, Conceptualization. **Lei Fan:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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